



PHASE 1 · PRODUCTION RELEASE

BMW Automation Studio

Operations & User Handbook

Strategic business specifications and standard operating procedures for OMC leadership, designers, and QA teams — how the studio works, and the productivity it unlocks.

Version 1.0

Production Release

Platform

macOS 12.0+

Audience

OMC, Design & QA

 DOCUMENT CONTROL

Document Overview

Document Title	BMW Automation Studio — Operations & User Handbook
Version	1.0 (Production Release)
Target Audience	OMC Leadership, Account Directors, Creative Directors, Production Operations Managers, Art Directors, Production Designers, QA Engineers & Proofreaders
Applicable Platforms	macOS 12.0+ (Apple Silicon & Intel)
Core Ecosystem	Adobe Illustrator, Adobe Photoshop, Apple Vision Framework, Electron, React
Confidentiality	Proprietary & Confidential — OMC & BMW Group

HOW TO USE THIS HANDBOOK

This handbook covers **Phase 1** of the BMW Automation Studio: automated dealer artwork adaptation for designers, and computer-vision-driven QA evaluation for proofreaders. Section 1–2 make the business case; Section 3 explains how the system is built; Sections 4–5 are step-by-step operating guides for designers and QA engineers respectively; Sections 6–7 cover maintenance and troubleshooting.

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SECTION 01

Executive Summary & OMC Business Context

1.1 The Role of the OMC in Campaign Rollouts

The **Omnichannel Marketing Center (OMC)** serves as the operational engine for BMW marketing. Operating within the creative production agency framework, the OMC is tasked with:

- **Centralizing Marketing Production** — taking central master creative concepts from global and national creative agencies and localizing them across regional hubs.
- **Managing Multi-Channel Distribution** — generating high-fidelity visual assets across Print, Out-of-Home, Digital Display, Social Media, and CRM / Direct Marketing.
- **Managing Dealer Syndication** — ensuring localized regional dealer networks receive compliant, tailored marketing collateral featuring exact legal trade names, addresses, phone numbers, and digital properties.

1.2 The Production Paradox: The Manual Adaptation Bottleneck

Modern automotive marketing demands personalized, localized messaging at high scale. Traditional creative adaptation workflows, however, rely on manual, repetitive human labor:

WHERE THE TRADITIONAL WORKFLOW BREAKS DOWN

1 The Copy-Paste Trap

A designer opens a master .ai / .psd file, manually copies address and phone details from a spreadsheet, pastes them into the artwork, adjusts text frames, re-exports and renames each file. For 5 sizes across 12 dealers, that's 60 individual manual edits.

2 Human Fatigue & Transcription Errors

After hours of repetitive edits, typos creep into critical fields — a transposed digit in a phone number, a misspelled URL, an omitted regulatory disclaimer.

3 The Proofreading Bottleneck

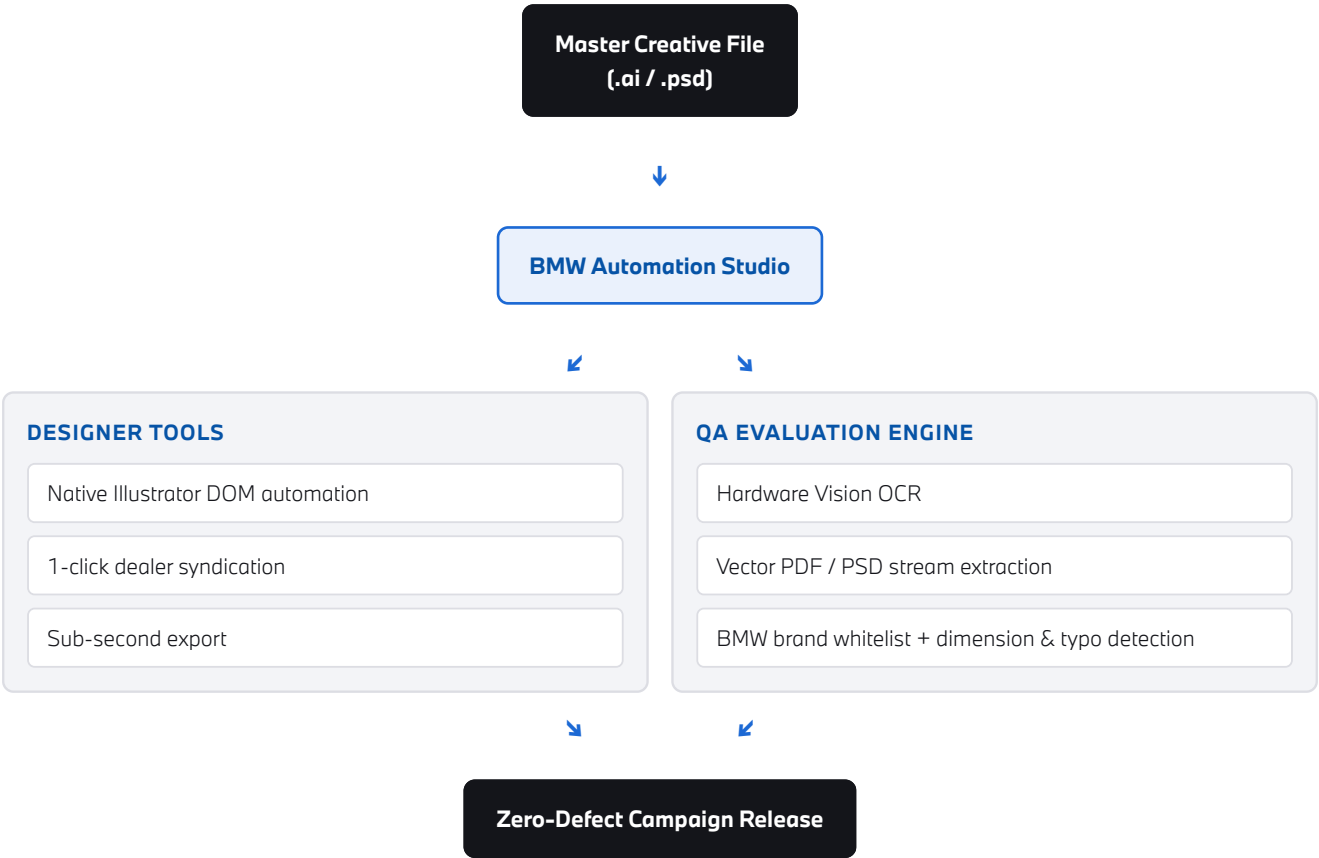
QA and copy-proofreaders must open each exported file one by one, manually reading microscopic 6pt disclaimers and cross-referencing contact sheets.

4 Catastrophic Failure Costs

A single printing error on a nationwide insert or roadside billboard can mean hundreds of thousands of dollars in reprints, regulatory scrutiny, or brand embarrassment.

1.3 Strategic Solution: End-to-End Automation

BMW Automation Studio bridges desktop creative software with native system intelligence. By unifying Adobe Illustrator scripting, Photoshop artboard parsing, Apple Vision neural OCR, and automated dictionary checking in a single native macOS application, the studio creates a closed-loop production environment:



SECTION 02

OMC Value Proposition & Operational Impact

2.1 Quantitative KPI Impact Matrix

OPERATIONAL METRIC	TRADITIONAL MANUAL WORKFLOW	WITH BMW AUTOMATION STUDIO	VARIANCE / BENEFIT
Adaptation Speed per Dealer	10 to 15 minutes	< 3 seconds	99% reduction
Turnaround Time (5 sizes x 10 dealers)	8.5 to 12.5 production hours	< 15 minutes (setup + export)	~97% reduction
Quality Control Inspection Time	3 to 5 minutes per asset	Instantaneous batch scan	95% reduction
Batch Archive QA (100 assets in ZIP)	5 to 6 hours of manual review	< 60 seconds automated scan	98% time saved
Transcription Error Escape Rate	3.5% to 5.0%	0.0% (automated binding)	Total elimination
Campaign Go-Live SLA	3 to 5 business days	Same-day (< 4 hours)	70–80% compression
Production Overtime & Remake Cost	High (frequent sprint overruns)	Near zero	Direct margin expansion

2.2 Core Operational Benefits for OMC

1 Throughput & Scalability Without Linear Headcount

A 40% increase in campaign volume traditionally required hiring more production artists or expensive freelance support. With automated batch variation generation, existing studio staff generate and inspect hundreds of assets in minutes — the OMC absorbs campaign surges effortlessly.

2 SLA Compression & Market Responsiveness

Automotive marketing is highly time-sensitive: seasonal campaigns, weekend test-drive events, and competitive rate adjustments demand rapid turnaround. Compressing the cycle from days to hours gives regional marketing directors unprecedented agility.

3 Zero-Defect Brand Integrity & Legal Governance

BMW enforces strict Corporate Identity guidelines: typography must use official BMW Type Next fonts, trademark capitalization must be exact (xDrive, sDrive, TwinPower Turbo, Steptronic, i4, i7, M Performance), and financial disclaimers (MSRP, APR, EMI, WLTP) must be present and legible. Automated validation checks every character against the official brand whitelist.

4 Relieving Creative Burnout & Raising Studio Morale

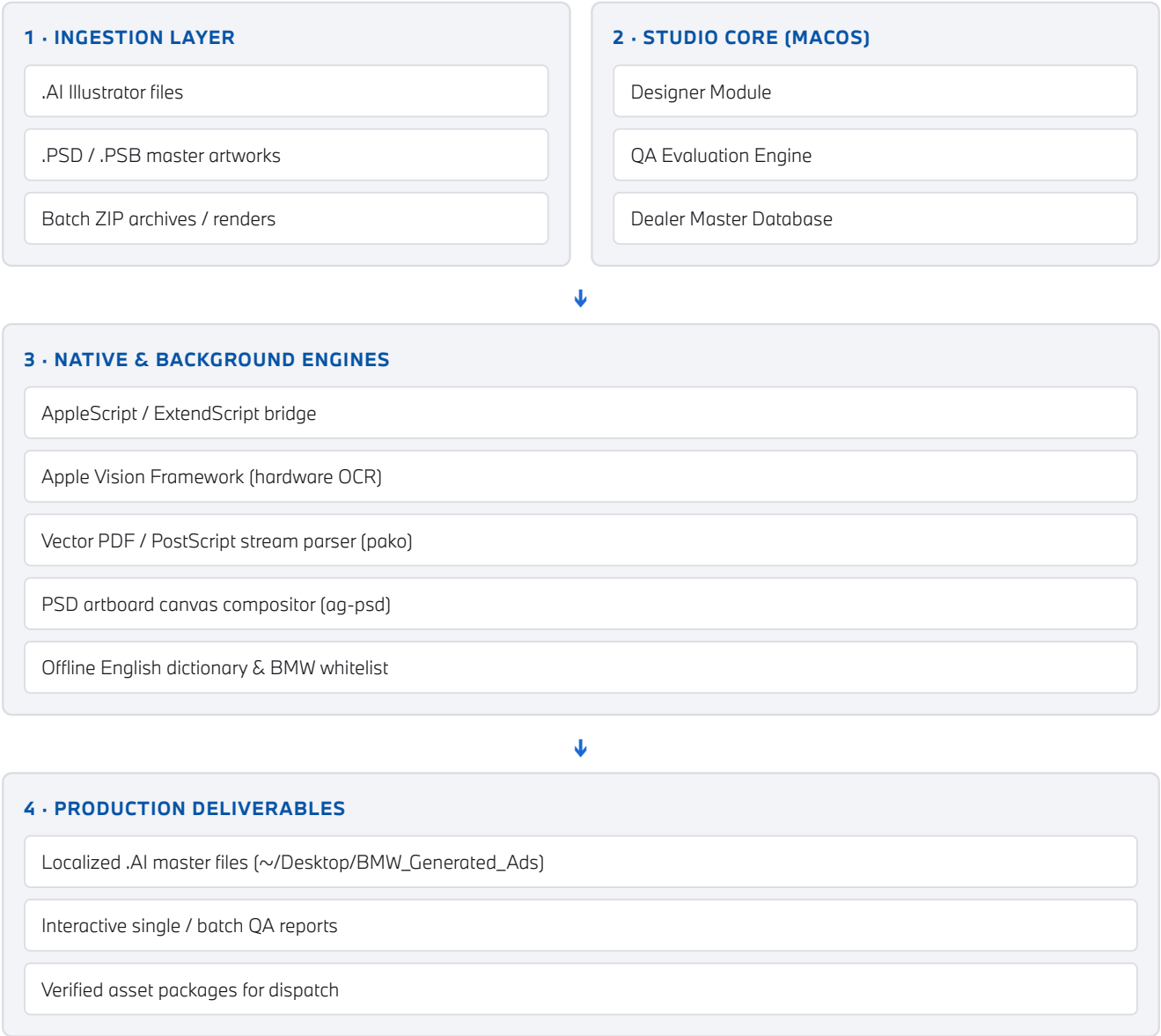
Manual copy-pasting and visual proofreading of disclaimer copy is mechanical, fatiguing work. Automating it elevates designers to focus on high-value creative adaptation, key visual ideation, and layout polish.

SECTION 03

System Architecture & Capabilities Overview

3.1 High-Level Operational Architecture

Assets flow from ingestion through the studio core, into native background engines, and out as production-ready deliverables:



3.2 The Three Operational Pillars

DESIGNER TOOLS	DEVELOPER TOOLS	QA EVALUATION
• Master ingestion • Live AI sync • SVG overlay map • Batch variations	• BMW EDM scaffolding • Email HTML compliance • Responsive templates • Brand layout guards	• Universal multi-format • Apple Vision OCR • Deep artboard inspection • BMW whitelist check

Designer Tools focuses on upstream production — extracting visual layers from Illustrator, binding target dealer text frames, and batch-exporting artwork without human transcription. **Developer Tools** standardizes Email Direct Marketing generation, enforcing BMW digital standards across HTML email layouts. **QA Evaluation** focuses on downstream verification — scanning final assets, whether exported by the tool or submitted by external agency partners, for dimensions, aspect ratios, file specs, text accuracy, and brand nomenclature.

SECTION 04

Designer's Operational Guide & Technical SOP

4.1 Workstation Setup & Prerequisites

Before running the Designer Tools module, confirm your macOS workstation meets these operational criteria:

PREREQUISITES CHECKLIST

- 1 Operating System**
macOS Monterey (12.0) or higher — Sonoma or Sequoia recommended.
- 2 Adobe Illustrator**
Installed and licensed via Adobe Creative Cloud.
- 3 macOS Automation Permissions**
On first launch, macOS will prompt "BMW Auto wants access to control Adobe Illustrator" — click Allow. If declined by mistake, enable it manually under
System Settings > Privacy & Security > Automation > BMW Auto > Enable Adobe Illustrator.
- 4 BMW Corporate Fonts**
Ensure BMWTypeNext-Light.otf and BMWTypeNext-Bold.otf are installed in macOS Font Book.

4.2 Master Artwork Preparation Standards

To guarantee 100% automated synchronization between Illustrator and the Studio, designers must follow these master setup rules:

MASTER ARTWORK PREPARATION RULES

- 1 Text Layer Isolation**
Keep the dealer address, phone number, and URL in a dedicated, un-grouped text frame.
- 2 Unlocked Status**
Ensure the target text frame and its parent layer are unlocked and visible in the Layers panel.
- 3 Enable PDF Compatibility on Save**
Always save with "Create PDF Compatible File" checked.
- 4 Single Master Artboard**
Ensure the active master layout sits on Artboard 1.

WHY PDF COMPATIBILITY IS MANDATORY

When an .ai file is saved with PDF compatibility enabled, Illustrator embeds a pristine vector PDF stream inside the file container. The Studio uses this stream to generate crisp, high-DPI in-app previews without launching Illustrator. Without it, the Studio falls back to low-resolution XMP thumbnails.

4.3 Step-by-Step Workflow: Automated Dealer Adaptations

1 Open the master in Adobe Illustrator

Launch Illustrator and open the master campaign artwork (e.g. `BMW_X5_Launch_Print_Master.ai`). Keep the document open.

2 Select Designer Tools in BMW Automation Studio

Launch BMW Auto from Applications. In the left sidebar, select **Designer Tools**.

3 Ingest the master design file

Under Process Master File, click Select File and pick the .ai or .psd master. The system analyzes it and shows a high-resolution preview plus metadata (dimensions, size, format).

4 Sync layers from Adobe Illustrator

Click "Sync Layers from Illustrator." The app runs an AppleScript bridge query against the live document, calculates artboard boundaries, reads all visible text frames, and overlays interactive blue bounding boxes across every text element in the preview.

5 Bind the target dealer layer

Hover over text boxes in the preview — hovering illuminates a block. Click directly on the dealer address block; it turns bright green and the status updates to "Target text block selected."

6 Select dealerships from the database

In the BMW Dealers Database panel, check the boxes for every dealership required for this adaptation sprint. Individual or multiple locations can be selected.

7 Execute batch generation

Click "Generate AI Variations." Behind the scenes, the engine communicates with the live Illustrator session, preserves the original text block state, iterates through every selected dealer, injects the formatted details (name, tel, location, URL), executes `doc.saveAs` with full PDF compatibility, and reverts the master document to its pristine original state on completion.

```
Saved to: ~/Desktop/BMW_Generated_Ads/<dealer_name>.ai
```

8 Verification alert

A confirmation alert reads "Successfully generated X variations! Saved to: Desktop/BMW_Generated_Ads/." Click OK and open the desktop folder to inspect the production-ready assets.

4.4 Best Practices & Common Pitfalls for Designers

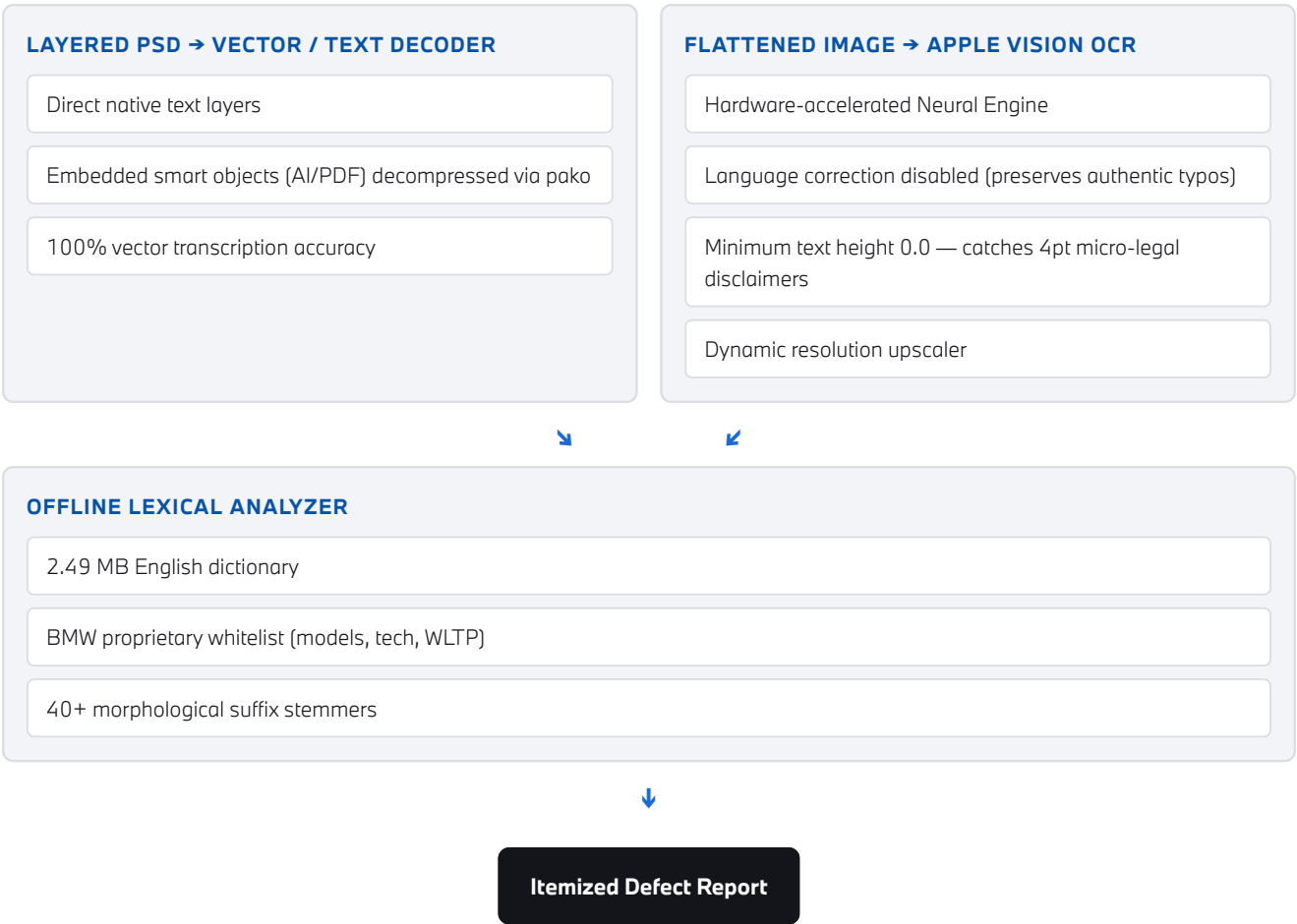
SCENARIO	PROBLEM	CORRECTIVE ACTION
"No documents open in Illustrator"	The master file was closed in Illustrator.	Re-open the master file in Illustrator before clicking "Sync Layers."
"Could not find target layer"	Target layer was locked or deleted after syncing.	Unlock the layer in Illustrator's Layers panel and click "Sync Layers" again.
Preview looks blurry / pixelated	File saved without PDF compatibility.	In Illustrator, File > Save As, check "Create PDF Compatible File," and re-select the file in the Studio.
Text frame overflows in output	Replacement address has longer lines than the master text box.	Set the text frame to Area Type with sufficient vertical breathing room rather than fixed Point Type.

SECTION 05

QA Engineer & Proofreader's Operational Guide & Technical SOP

5.1 The QA Philosophy: Hybrid Deterministic & Computer Vision QC

Traditional OCR tools fail in automotive advertising proofreading for two reasons: standard engines auto-correct typos with dictionary language models — hiding real errors — and standard web OCR struggles with 300 DPI print files or micro-legal disclaimers on photographic backgrounds.



5.2 Asset Ingestion Standards

MODE 1

Single Asset

Drag and drop any individual graphic: .jpg, .jpeg, .png, .webp, .bmp, .tiff. Used for quick ad-hoc inspection of a master key visual or social post.

MODE 2

Multi-Artboard Master PSD

Drag and drop a layered .psd / .psb. The engine auto-detects every embedded artboard (Artboard 1, 300×250, Instagram_Story_9×16, etc.) and evaluates each as an independent sub-asset with dedicated previews and metadata.

MODE 3

Batch ZIP Archive

Drag and drop a .zip containing tens or hundreds of exported banners. The engine decompresses in-memory, strips macOS hidden folders ().__MACOSX, .DS_Store), and evaluates every contained image sequentially.

5.3 Step-by-Step Workflow: Ingestion, Analysis & Evaluation

1

Navigate to QA Evaluation

Click QA Evaluation in the left sidebar to open the file dropzone.

2

Ingest assets

Drag and drop the asset file (e.g. `BMW_Summer_Campaign_Assets.zip` or individual .png/.psd files) onto the dropzone, or click to browse via Finder.

3

Monitor ingestion & analysis progress

A real-time progress bar cycles through: reading PSD file buffer, parsing layers & artboards, decompressing embedded smart objects, running OCR analysis (asset X of Y), checking spelling against the BMW whitelist.

4

Access the QC Report

Once analysis reaches 100%, the Studio immediately presents the QC Report.

5.4 Interpreting the QC Report

The report dynamically switches between Single Image Mode and Batch Report Mode.

ImagePreview

Width × Height

1920 × 1080 px

Aspect Ratio / Color

16:9 · 24-bit RGB

Format / Size

JPEG · 1.42 MB

Status

2 Issues

OCR Confidence

88%

"The all-new BMW i7 with xDrive... Contact us for special financng terms"

Flagged Spelling Issue

financng

Dimensions & Specs card confirms canvas pixel width, height, aspect ratio and color depth. **File Information card** shows format, formatted size and last-modified timestamp. **Content & Spelling card** shows an OCR confidence meter (100% for native PSD text), the complete verbatim extracted text, and every flagged word highlighted in red with surrounding context for instant diagnosis.

In **Batch Report Mode**, KPI cards at the top show Total Images, Total Size, Passed, and Flagged counts, with an interactive Asset Table below (Thumbnail, File Name, Dimensions, Format & Size, Status) — clicking any row expands the full inspection drawer without leaving the batch view.

5.5 Defect Classification & Decision Matrix

STATUS BADGE	TRIGGER CONDITION	SEVERITY	REQUIRED QA ACTION
Flagged	Word detected that is neither in the English dictionary nor the BMW whitelist (e.g. "financng," "exclusiv").	High (Defect)	Reject asset. Log the exact typo and notify the production artist for immediate correction.
Pass	Text extracted cleanly; all words match standard dictionary or verified BMW nomenclature.	None (Approved)	Approve asset. File passes content quality control.
— No Text	No text glyphs detected (pure photographic backgrounds, textures, or image-only crops).	Informational	Verify intent. If disclaimers were expected, flag as missing copy; if purely background art, approve.
False Positive	Valid local proper noun, new dealership name, or regional city not yet in the whitelist (e.g. new dealer "Bavaria Motors").	Low	Verify & log. Confirm manually; if accurate, request the Tech Lead add the word to the BMW whitelist.

SECTION 06

Technical Maintenance & Customization

6.1 Updating Dealership Master Records

The dealership directory is maintained in `src/components/DealersDatabase.tsx`.

- 1 Open `DealersDatabase.tsx`.

- 2 Locate the `MOCK_DEALERS` array and add new dealership entries:

```
export const MOCK_DEALERS = [  
  { id: '1', name: 'BMW Infinity Cars', location: 'Worli', tel: '+91 22 67145100', url: 'www.bmw-infinitycars.com' },  
  { id: '2', name: 'BMW Navnit Motors', location: 'Andheri', tel: '+91 22 66777777', url: 'www.bmw-navnitmotors.com' },  
  // Add new dealership entries here:  
  { id: '6', name: 'BMW Deutsche Motoren', location: 'Delhi NCR', tel: '+91 11 41000000', url: 'www.bmw-deutschemotoren.com' },  
];
```

- 3 Save the file. In dev mode, Vite hot-reloads immediately.

6.2 Maintaining the BMW Brand Whitelist & Dictionary

The BMW-specific terminology whitelist resides in `src/components/qaUtils.ts` under `BMW_WHITELIST`.

- 1 Open `qaUtils.ts`.

- 2 Add lowercase strings to the whitelist `Set`:

```
const BMW_WHITELIST = new Set([  
  // New 2026 Models  
  'neue', 'klasse', 'ix7', 'm3cs',  
  
  // New Technologies  
  'panoramic', 'vision', 'bidirectional',  
  
  // Regional / Market specifics  
  'delhi', 'bengaluru', 'mumbai', 'pune',  
]);
```

- 3 Save the file — the QA spell checker immediately recognizes the additions without flagging them.

6.3 Recompiling the Native OCR Engine

The native OCR engine is written in Swift (`electron/ocr-vision.swift`) and compiled into a standalone Darwin Mach-O binary (`electron/ocr-vision`). If the Swift source is modified — for example, adjusting the recognition level or text detection heuristics:

```
# Navigate to the workspace root
cd bmw-app/bmw-macos-app

# Compile for Apple Silicon (M1/M2/M3/M4):
swiftc -O -target arm64-apple-macos12.0 electron/ocr-vision.swift -o electron/ocr-vision

# Ensure binary execution permissions:
chmod +x electron/ocr-vision
```

SECTION 07

Operational Troubleshooting & FAQs

Q1 When clicking "Sync Layers," the app hangs or alerts "AppleScript Error."**ROOT CAUSE**

Adobe Illustrator is unresponsive or showing an open dialog (Missing Font, Color Profile Mismatch, Update Links), or macOS blocked automation.

RESOLUTION

Switch to Illustrator and dismiss any modal dialogs, then verify BMW Auto is permitted to control Illustrator under **System Settings > Privacy & Security > Automation**.

Q2 Designer Tools generated variations — where did the files go?**RESOLUTION**

All generated variations are automatically organized on the active user's desktop: `~/Desktop/BMW_Generated_Ads/<dealer_name>.ai`.

Q3 Why does a 300×250 banner ad fail to detect small 5pt disclaimer text?**ROOT CAUSE**

Low pixel resolution on standard digital banners — where 5pt text may only be 4–6 pixels high — can fall beneath standard optical OCR thresholds.

RESOLUTION

The engine includes an automatic 4× pre-processing upscaler for images under 1400px. If misses persist on raster assets, drop the master PSD instead — the Studio extracts vector text directly from the PSD layer tree with 100% mathematical fidelity.

Q4 How do I perform a complete reset/reload of the Studio mid-session?**RESOLUTION**

Click "Refresh App" in the top window bar or bottom of the sidebar, or press `Cmd + R` inside the window to reload the web view without closing the background Electron process.